

Large Foundation Models: Diffusion, Flow Matching, and AI for Science

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What I cannot create, I do not understand.

Outline

- 1 Preliminary: Generating Samples from Probability Distributions
- 2 Matching score for training
- 3 Relations of different models: x_0 , epsilon and score model
- 4 Potential Score Matching loss
- 5 Diffusion model: from theories to implementation
- 6 Continuous version of Diffusion Model
- 7 Mathematical motivation for Flow Matching
- 8 Connection with Diffusion Models
- 9 Conditional Paths and Velocity Targets
- 10 Flow Matching Training and Inference

Generate samples from given distribution

Given a distribution $p(x)$, how to sample from the distribution?

- Uniform distribution & Gaussian distribution: easy sampling
- Mixture Gaussian: determine which Gaussian to sample & sample from that Gaussian
- A general sample function?
 - Langevin Markov Chain Monte Carlo sampling (Langevin MCMC)
 - Stein Variational Gradient Descent (SVGD)
 - ...

Score function is all you need?

A typical process of generating new samples from modeling the data distribution

- ① To find the probability function $p(x)$, model $p(x; \theta)$.
- ② Normalization distribution $p(x; \theta) = \frac{1}{C(\theta)} e^{-E(x; \theta)}$ with $C(\theta) \equiv \int_x e^{-E(x; \theta)} dx$.
- ③ Generating samples from above sampling methods.

Recall the Langevin Markov Chain Monte Carlo sampling

$$x_t = x_{t-1} + \frac{\Delta t}{2} \nabla_x \log p(x_{t-1}; \theta) + \sqrt{\Delta t} \epsilon, \quad \epsilon \sim N(0, 1). \quad (1)$$

Here $\nabla_x \log p(x_{t-1}; \theta) = -\nabla_x E(x; \theta)$ is free of the normalization $C(\theta)$.

Define the score function

$$S(x; \theta) \equiv \nabla \log p(x; \theta). \quad (2)$$

More about Langevin MCMC and the score function

Continuous form of the Langevin MCMC

- $\Delta t \rightarrow 0$

$$dx = \frac{1}{2} \nabla_x \log p(x) dt + dB_t, \quad (3)$$

where dB_t is a Brownian motion.

- Given $p(x) = \frac{1}{C} e^{-E}$, we have

$$dx = -\frac{1}{2} \nabla_x E dt + dB_t \quad (4)$$

Langevin MCMC is overdamped Langevin dynamics

- Langevin dynamics (MD)

$$\ddot{x} = -\nabla_x E - \gamma \dot{x} + dB_t, \quad (5)$$

- Given $\gamma \dot{x} \gg \ddot{x}$, we have the overdamped Langevin equation

$$dx = -\frac{1}{\gamma} \nabla_x E + \frac{1}{\gamma} dB_t.$$

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Matching the score by neural networks

Recall the definition

$$S \equiv \nabla_x \log p(x), \quad (7)$$

The key is to matching the score by a neural network

$$J_{ESM} = \frac{1}{2} \mathbb{E}_p \left[\left\| s(x; \theta) - \frac{\partial \log p(x)}{\partial x} \right\|^2 \right], \quad (8)$$

where J_{ESM} is called Explicit Score Matching loss. However, it is not tractable to compute $\frac{\partial \log p(x)}{\partial x}$ from data.

Where should we go?

Implicit Score Matching (ISM) loss

Implicit Score Matching loss (Aapo, 2005) was developed to make a tractable score matching

$$J_{ISM} \equiv \mathbb{E}_p \left[\frac{1}{2} \|s(x; \theta)\|^2 + \nabla \cdot s(x; \theta) \right] = J_{ESM} - C. \quad (9)$$

ISM Proof: Expansion of ESM

Proof.

First, we expand the ESM loss:

$$J_{ESM} = \frac{1}{2} \mathbb{E}_p \left[\left\| s(x; \theta) - \frac{\partial \log p(x)}{\partial x} \right\|^2 \right] \quad (10)$$

$$= \frac{1}{2} \mathbb{E}_p [\|s\|^2] - \mathbb{E}_p \left[s \cdot \frac{\partial \log p}{\partial x} \right] + \frac{1}{2} \mathbb{E}_p \left[\left\| \frac{\partial \log p}{\partial x} \right\|^2 \right] \quad (11)$$

The key insight is to simplify the cross term using integration by parts. □

ISM Proof: Integration by Parts

Proof (Cont).

The cross term can be simplified:

$$-\mathbb{E}_p[s \cdot \frac{\partial \log p}{\partial x}] = - \int_x ps \cdot \nabla \log p dx \quad (12)$$

$$= - \int_x ps \cdot \frac{\nabla p}{p} dx = - \int_x s \cdot \nabla p dx \quad (13)$$

$$= - \int \nabla \cdot (ps) dx + \int p \nabla \cdot s dx = \mathbb{E}_p[\nabla \cdot s] \quad (14)$$

Proof (Cont).

Therefore, we have:

$$J_{ISM} = J_{ESM} - C \quad (15)$$

where $C \equiv \frac{1}{2} \mathbb{E}_p \left[\left\| \frac{\partial \log p}{\partial x} \right\|^2 \right]$ is a constant independent of θ .

This makes ISM tractable since we don't need to compute the true score function.

Denoising Score Matching (DSM) loss

However, the ISM score is not very stable to optimize, with two reasons

- ① Expectation is performed on the whole distribution;
- ② The loss is negative decreasing to $-C$ with a great quantity, e.g. $-1e5$.

Denoising Score Matching (DSM) loss (Vincent, 2011) is developed to solve this problem

$$J_{DSM} = \frac{1}{2} \mathbb{E}_{p(x, \tilde{x})} [\|s(\tilde{x}; \theta) - \nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2]. \quad (16)$$

And we can prove that $J_{DSM} = J_{ESM} + C$.

DSM Proof: Expansion

Proof.

We expand the DSM loss:

$$J_{DSM} \equiv \frac{1}{2} \mathbb{E}_{p(x, \tilde{x})} [\|s(\tilde{x}; \theta) - \nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2] \quad (17)$$

$$= \frac{1}{2} \mathbb{E}_{p(\tilde{x})} [\|s(\tilde{x})\|^2] - \mathbb{E}_{p(x, \tilde{x})} [s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x}|x)] \quad (18)$$

$$+ \frac{1}{2} \mathbb{E}_{p(x, \tilde{x})} [\|\nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2] \quad (19)$$

The key step is to show that the cross terms of J_{ESM} and J_{DSM} are equal. □

DSM Proof: Cross Term Equivalence

Proof (Cont).

We need to show:

$$\mathbb{E}_p(\tilde{x})[s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x})] = \mathbb{E}_{p(\tilde{x}, x)}[s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x}|x)] \quad (20)$$

Starting from the left side:

$$= \int_{\tilde{x}} p(\tilde{x}) s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x}) d\tilde{x} \quad (21)$$

$$= \int_{\tilde{x}} s(\tilde{x}) \cdot \nabla_{\tilde{x}} p(\tilde{x}) d\tilde{x} \quad (22)$$

DSM Proof: Cross Term Equivalence (Cont)

Proof (Cont).

Continuing the derivation:

$$= \int_{\tilde{x}} \int_x p(x) s(\tilde{x}) \cdot \nabla_{\tilde{x}} p(\tilde{x}|x) d\tilde{x} dx \quad (23)$$

$$= \int_{\tilde{x}} \int_x p(\tilde{x}|x) p(x) s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x}|x) d\tilde{x} dx \quad (24)$$

$$= \mathbb{E}_{p(\tilde{x}, x)} [s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x}|x)] \quad (25)$$

This completes the proof of cross term equivalence.

Proof (Cont).

Therefore, we have:

$$J_{DSM} = J_{ESM} - \frac{1}{2} \mathbb{E}_p [\|\nabla_{\tilde{x}} \log p(\tilde{x})\|^2] + \frac{1}{2} \mathbb{E}_{p(x, \tilde{x})} [\|\nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2] \quad (26)$$

Since the last two terms are constants independent of θ , we have $J_{DSM} = J_{ESM} + C$ where C is a constant.

DSM is all you need?

Problem arises when $\tilde{x} \rightarrow x$. We can prove that when Gaussian noise added by $p(\tilde{x}|x) = N(\alpha x, \sigma^2)$,

$$J_{DSM} \rightarrow \infty, \text{ if } \tilde{x} \rightarrow x. \quad (27)$$

Proof.

Given the Gaussian noise added, we have

$$\nabla_{\tilde{x}} \log p(\tilde{x}|x) = \nabla_{\tilde{x}} \log e^{-\frac{(\tilde{x}-\alpha x)^2}{2\sigma^2}} = -\frac{\tilde{x} - \alpha x}{\sigma^2}. \quad (28)$$

And we can show that the following formula goes to infinity

$$\begin{aligned} \mathbb{E}_{p(x, \tilde{x})} [\|\nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2] &= \mathbb{E}_{p(x, \tilde{x})} [\|\frac{\tilde{x} - \alpha x}{\sigma^2}\|^2] \\ &= \mathbb{E}_{p(x)} \mathbb{E}_{p(\tilde{x}|x)} [\|\frac{\tilde{x} - \alpha x}{\sigma^2}\|^2] = \mathbb{E}_{\epsilon \sim N(0,1)} [\|\frac{\epsilon}{\sigma}\|^2] \rightarrow \infty \text{ if } \sigma \rightarrow 0. \end{aligned} \quad (29)$$

Finally, we know that

$$J_{DSM} = J_{ESM} + C \rightarrow \infty \text{ as } C \rightarrow \infty \text{ when } \tilde{x} \rightarrow x. \quad (30)$$

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Revisit the adding noise from x to $\tilde{x}$

Gaussian transition kernel for adding noise from x to $\tilde{x}$

$$\tilde{x} = \alpha x + \sigma \epsilon, \quad (31)$$

which equivalent to the Gaussian transition kernel

$$p(\tilde{x}|x) = N(\tilde{x}; \alpha x, \sigma^2), \quad (32)$$

where $\epsilon \sim N(0, 1)$ and

$$N(\tilde{x}; \alpha x, \sigma^2) = C e^{-\frac{\|\tilde{x} - \alpha x\|^2}{2\sigma^2}}. \quad (33)$$

And

$$\epsilon = \frac{\tilde{x} - \alpha x}{\sigma}. \quad (34)$$

Questions arise: how can we get x given $\tilde{x}$?

Tweedie's Formula

Tweedie's Formula (Bayes statistics) states (Robbins, 1956)

Theorem

Given random variables x, y , $y \sim N(x, \sigma^2 I)$, i.e., $p(y|x) = N(x, \sigma^2)$, the expectation of x could be given by

$$\mathbb{E}[x|y] = y + \sigma^2 \nabla \log p(y). \quad (35)$$

Let

$$x \leftarrow \alpha x, \quad y \leftarrow \tilde{x}, \quad (36)$$

we have

$$s \equiv \nabla_{\tilde{x}} \log p(\tilde{x}) = -\frac{\tilde{x} - \alpha \mathbb{E}[x|\tilde{x}]}{\sigma^2} = -\frac{1}{\sigma} \mathbb{E}\left[\frac{\tilde{x} - \alpha x}{\sigma} | \tilde{x}\right] = -\frac{\mathbb{E}[\epsilon|\tilde{x}]}{\sigma}, \quad (37)$$

where we have used the fact $\epsilon = \frac{\tilde{x} - \alpha x}{\sigma}$.

Another insight: from the definition of the score S

Recall that adding noise by $\tilde{x} = \alpha x + \sigma \epsilon$,

$$p(\tilde{x}|x) = Ce^{-\frac{\|\tilde{x}-\alpha x\|^2}{2\sigma^2}}. \quad (38)$$

Hence

$$s(\tilde{x}) = \nabla_{\tilde{x}} \log p(\tilde{x}) = \frac{\nabla_{\tilde{x}} p(\tilde{x})}{p(\tilde{x})} = \frac{\int_x \nabla_{\tilde{x}} p(\tilde{x}|x) p(x) dx}{p(\tilde{x})}, \quad (39)$$

From Eq. (38), we have

$$s(\tilde{x}) = \frac{\int_x \nabla_{\tilde{x}} \log p(\tilde{x}|x) p(\tilde{x}|x) p(x) dx}{p(\tilde{x})} = \int_x \nabla_{\tilde{x}} \log p(\tilde{x}|x) p(x|\tilde{x}) dx, \quad (40)$$

leading to

$$s(\tilde{x}) = \mathbb{E}_x[\nabla_{\tilde{x}} \log p(\tilde{x}|x)|\tilde{x}] = \mathbb{E}_x[-\frac{\tilde{x} - \alpha x}{\sigma^2}|\tilde{x}] = -\frac{\tilde{x} - \alpha \mathbb{E}[x|\tilde{x}]}{\sigma^2}.$$

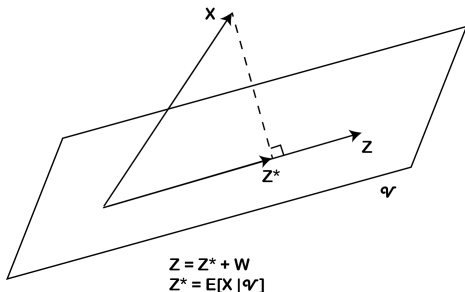
Theoretical Support: Conditional Expectation

Theorem

Let X be an integrable random variable. Then for each σ -algebra $\mathcal{V}$ and $Z \in \mathcal{V}$, $Z^* = \mathbb{E}(X|\mathcal{V})$ solves the least square problem

$$\operatorname{argmin}_{Z \in \mathcal{V}} \|Z - X\|,$$

where $\|Z\| = \left(\int Z^2 dP\right)^{\frac{1}{2}}$.



Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $L^2 := L^2(\Omega, \mathcal{F}, \mathbb{P})$ with $\langle U, V \rangle = \mathbb{E}[UV]$. Fix a sub- σ -field $\mathcal{G} \subset \mathcal{F}$ and denote the closed subspace

$$\mathcal{V} := L^2(\mathcal{G}) = \{Z \in L^2 : Z \text{ is } \mathcal{G}\text{-measurable}\}.$$

Claim. For $X \in L^2$,

$$Z^* := \mathbb{E}[X \mid \mathcal{G}] \in \mathcal{V} \quad \text{and} \quad Z^* = \arg \min_{Z \in \mathcal{V}} \mathbb{E}[(Z - X)^2].$$

Proof of the claim

For any $Z \in \mathcal{V}$,

$$\langle X - Z^*, Z \rangle = \mathbb{E}[(X - Z^*)Z] = \mathbb{E}(\mathbb{E}[(X - Z^*)Z \mid \mathcal{G}]) = \mathbb{E}(Z \mathbb{E}[X - Z^* \mid \mathcal{G}]) = 0.$$

Hence $X - Z^* \perp \mathcal{V}$.

Let any $Z \in \mathcal{V}$ be written as $Z = Z^* + W$ with $W \in \mathcal{V}$. Then

$$\|X - Z\|_2^2 = \|X - (Z^* + W)\|_2^2 = \|X - Z^*\|_2^2 + \|W\|_2^2 \geq \|X - Z^*\|_2^2,$$

because $\langle X - Z^*, W \rangle = 0$. Equality $\Leftrightarrow W = 0 \Leftrightarrow Z = Z^*$, proving uniqueness and the minimizer claim.

Revisit several models

① Score model

$$Loss_{score} = \|s_{\theta}(\tilde{x}) - \frac{\epsilon}{\sigma}\|^2; \quad (42)$$

② X prediction model (denoising model, x_0 model)

$$Loss_{x_0} = \|x_{\theta}(\tilde{x}) - x\|^2, \quad (43)$$

with

$$s(\tilde{x}) = -\frac{\tilde{x} - \alpha x_{\theta^*}(\tilde{x})}{\sigma^2} \quad (44)$$

where $x_{\theta}(\tilde{x}) \rightarrow x_{\theta^*}(\tilde{x}) \equiv \mathbb{E}[x|\tilde{x}]$;

③ ϵ prediction model

$$Loss_{\epsilon} = \|\epsilon_{\theta}(\tilde{x}) - \epsilon\|^2, \quad (45)$$

with

$$s(\tilde{x}) = -\frac{\epsilon_{\theta^*}(\tilde{x})}{\sigma}, \quad (46)$$

where $\epsilon_{\theta}(\tilde{x}) \rightarrow \epsilon_{\theta^*}(\tilde{x}) \equiv \mathbb{E}[\epsilon|\tilde{x}]$.

The consistency of the three modeling

As stated before, we have the following connection in the optimal (θ^*) case

$$s_{\theta^*}(\tilde{x}) = -\frac{\tilde{x} - \alpha x_{\theta^*}(\tilde{x})}{\sigma^2} = -\frac{\epsilon_{\theta^*}(\tilde{x})}{\sigma}, \quad (47)$$

by the above conditional expectation.

Hence, in the real modeling of designing the loss, we use the connection of three models without reaching the optimal parameter

$$s_{\theta}(\tilde{x}) = -\frac{\tilde{x} - \alpha x_{\theta}(\tilde{x})}{\sigma^2} = -\frac{\epsilon_{\theta}(\tilde{x})}{\sigma}, \quad (48)$$

and substitute each other form to the loss definition above, we can recover all of the losses given by different models.

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Key Identity: Jacobian Switch (chain rule between $\mathbf{x}_0$ and $\mathbf{x}_t$)

The Gaussian transition density is

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \alpha_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}) \propto \exp\left(-\frac{\|\mathbf{x}_t - \alpha_t \mathbf{x}_0\|^2}{2\sigma_t^2}\right).$$

A basic but crucial identity:

$$\nabla_{\mathbf{x}_t} q(\mathbf{x}_t | \mathbf{x}_0) = -\frac{1}{\alpha_t} \nabla_{\mathbf{x}_0} q(\mathbf{x}_t | \mathbf{x}_0), \quad (\alpha_t > 0).$$

Proof sketch. Differentiating the exponent w.r.t. $\mathbf{x}_t$ and w.r.t. $\mathbf{x}_0$ shows

$\nabla_{\mathbf{x}_t} \|\mathbf{x}_t - \alpha_t \mathbf{x}_0\|^2 = 2(\mathbf{x}_t - \alpha_t \mathbf{x}_0)$, $\nabla_{\mathbf{x}_0} \|\mathbf{x}_t - \alpha_t \mathbf{x}_0\|^2 = -2\alpha_t(\mathbf{x}_t - \alpha_t \mathbf{x}_0)$, thus the factor $-1/\alpha_t$.

Theorem 1: Score as Conditional Force Expectation

Statement. With $p(\mathbf{x}_0) \propto e^{-E(\mathbf{x}_0)/k_B T}$ and $\mathbf{x}_t = \alpha_t \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$,

$$\boxed{\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) = \frac{1}{\alpha_t} \mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t} \left[\frac{\mathbf{F}(\mathbf{x}_0)}{k_B T} \right]}, \quad \mathbf{F}(\mathbf{x}_0) = -\nabla_{\mathbf{x}_0} E(\mathbf{x}_0).$$

Derivation.

$$\begin{aligned} \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) &= \frac{\nabla_{\mathbf{x}_t} \int p(\mathbf{x}_0) q(\mathbf{x}_t | \mathbf{x}_0) d\mathbf{x}_0}{p(\mathbf{x}_t)} \\ &= \frac{1}{p(\mathbf{x}_t)} \int p(\mathbf{x}_0) \nabla_{\mathbf{x}_t} q(\mathbf{x}_t | \mathbf{x}_0) d\mathbf{x}_0 \end{aligned}$$

Theorem 1: Derivation (Cont)

Proof (Cont).

Using the Jacobian switch identity (*):

$$\begin{aligned} &\stackrel{(*)}{=} \frac{1}{p(\mathbf{x}_t)} \int p(\mathbf{x}_0) \left(-\frac{1}{\alpha_t} \right) \nabla_{\mathbf{x}_0} q(\mathbf{x}_t | \mathbf{x}_0) d\mathbf{x}_0 \\ &= -\frac{1}{\alpha_t} \frac{1}{p(\mathbf{x}_t)} \int q(\mathbf{x}_t | \mathbf{x}_0) \nabla_{\mathbf{x}_0} p(\mathbf{x}_0) d\mathbf{x}_0 \end{aligned}$$

Theorem 1: Derivation (Final)

Proof (Cont).

$$\begin{aligned} &= -\frac{1}{\alpha_t} \int \underbrace{\frac{p(\mathbf{x}_0)q(\mathbf{x}_t | \mathbf{x}_0)}{p(\mathbf{x}_t)}}_{p(\mathbf{x}_0|\mathbf{x}_t)} \nabla_{\mathbf{x}_0} \log p(\mathbf{x}_0) d\mathbf{x}_0 \\ &= \frac{1}{\alpha_t} \mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t} \left[\frac{-\nabla_{\mathbf{x}_0} E(\mathbf{x}_0)}{k_B T} \right] = \frac{1}{\alpha_t} \mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t} \left[\frac{\mathbf{F}(\mathbf{x}_0)}{k_B T} \right] \end{aligned}$$

Experiments on first 1, 000 frames for LJ data

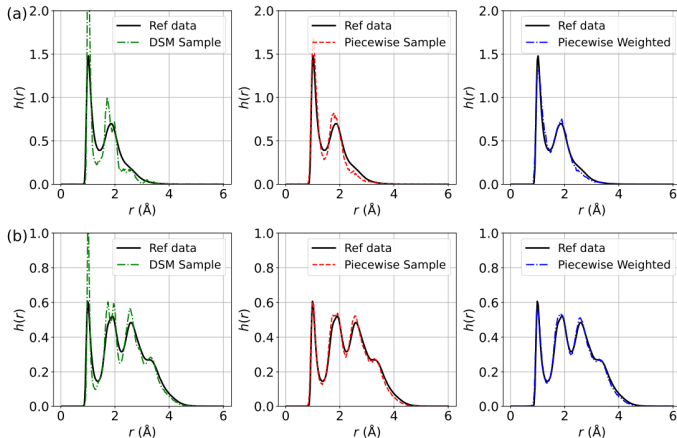


Figure 1: The distribution of interatomic distances $h(r)$ for LJ potential using biased training data with a sample size of 500. (a) Comparison of $h(r)$ for DSM, Piecwise, and Piecwise Weighted losses (from left to right) against reference data for LJ-13; (b) Comparison of $h(r)$ for DSM, Piecwise, and Piecwise Weighted losses for LJ-55.

Experiments on first 5, 000 frames for MD reference trajectories

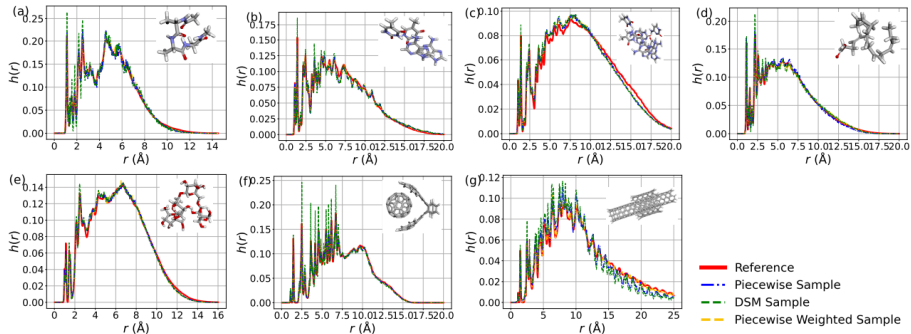


Figure 3: The distribution of interatomic distances ($r(\text{\AA})$) for (a) Ac-Ala3-NHMe, (b) DNA base pair (AT-AT), (c) DNA base pair (AT-AT-CG-CG), (d) Docosahexaenoic acid, (e) Stachyose, (f) Buckyball catcher, (g) Dw nanotube in MD22 dataset. The insets display the ball-and-stick representations of these molecules.

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From Theory to Practice

Now that we have established the theoretical foundations of score matching and diffusion processes, let's explore how these concepts are implemented in practice through several key approaches:

- **SMLD** (Score Matching Langevin Dynamics)
- **DDPM** (Denoising Diffusion Probabilistic Models)
- **Continuous formulations** and their advantages

Each approach offers different insights into the practical implementation of diffusion-based generative models.

Score Matching Langevin Dynamic (SMLD) (Song, 2019) adding noise in a following manner

$$p(\tilde{x}_i|x) = N(\tilde{x}_i; x, \sigma_i^2) \equiv Ce^{-\frac{\|\tilde{x}_i - x\|^2}{2\sigma_i^2}}, \quad (49)$$

where a geometric sequence $\sigma_1 > \sigma_2 > \dots > \sigma_T \approx 0$ is given to add noise with different level. In a random variable view,

$$\tilde{x}_i = x + \sigma_i \epsilon \quad (50)$$

for different σ_i to get different noised random variable $\tilde{x}_i$.

Training object J_{DSM} is given by

$$J_{DSM} = \frac{1}{2} \mathbb{E}_{\sigma_i} \mathbb{E}_{p(x)} \mathbb{E}_{p(\tilde{x}_i|x)} \left[\left\| s_{\theta}(\tilde{x}_i, \sigma_i) + \frac{\tilde{x}_i - x}{\sigma_i^2} \right\|^2 \right]. \quad (51)$$

Anneled Langevin MCMC for sampling

$$x_t = x_{t-1} + \frac{\Delta t}{2} S_{\theta}(x_{t-1}) + \sqrt{\Delta t} \epsilon, \quad \epsilon \sim N(0, 1).$$

denoising diffusion probabilistic models (DDPM) (Ho, 2020)

- ① Derived from the Evidence Lower BOund (ELBO), **Not** score matching.
- ② First provide adding noise and denoise in the forward and reverse process.

The main procedure

- ① Adding noise

$$x_t = \sqrt{1 - \beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon; \quad (53)$$

- ② Transition kernel

$$p(x_t|x_0) = N(x_t; \alpha_t x_0, \sigma_t^2), \quad (54)$$

where $\alpha_t = \prod_{i=1}^t \sqrt{1 - \beta_i}$ and $\sigma_t^2 = 1 - \alpha_t^2$.

The main procedure (Continued)

- 1 Adding noise

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon; \quad (55)$$

- 2 Transition kernel

$$p(x_t | x_0) = N(x_t; \alpha_t x_0, \sigma_t^2), \text{ i.e., } x_t = \alpha_t x_0 + \sigma_t \epsilon, \quad (56)$$

where $\alpha_t = \prod_{i=1}^t \sqrt{1 - \beta_i}$ and $\sigma_t^2 = 1 - \alpha_t^2$.

- 3 Training Loss

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{p(x)} \mathbb{E}_{p(x_t|x)} [\|\epsilon_\theta(x_t, t) - \epsilon\|^2], \quad (57)$$

where we have used the approximation $x_0 = \frac{x_t - \sigma_t \epsilon}{\alpha_t}$.

- 4 Sampling

$$x_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} (x_t + \beta_t s_\theta(x_t, t)) + \sqrt{\beta_t} \epsilon_t, \quad t = T, T-1, \dots, 1. \quad (58)$$

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The continuous version of SMLD

Recall that

$$x_t = x_0 + \sigma_t \epsilon, \quad (59)$$

we can prove that

$$x_t = x_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \epsilon. \quad (60)$$

Proof.

We can prove the formula by a Recurrence

$$\begin{aligned} x_t &= x_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \epsilon \\ &= x_{t-2} + \sqrt{\sigma_{t-1}^2 - \sigma_{t-2}^2} \epsilon + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \epsilon \\ &= x_{t-2} + \sqrt{\sigma_t^2 - \sigma_{t-2}^2} \epsilon \\ &= \dots = x_0 + \sigma_t \epsilon, \quad \text{where } \sigma_0 \approx 0. \end{aligned} \quad (61)$$

The continuous version of SMLD

From

$$x_t = x_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \epsilon, \quad (62)$$

we have

$$\begin{aligned} \Delta x_t &= \sqrt{\frac{\sigma_t^2 - \sigma_{t-1}^2}{\Delta t}} \sqrt{\Delta t} \epsilon \\ &= \sqrt{\frac{\sigma_t^2 - \sigma_{t-1}^2}{\Delta t}} \Delta B_t. \end{aligned} \quad (63)$$

Further, let $\Delta t \rightarrow 0$, we have

$$dx = \sqrt{\frac{d\sigma_t^2}{dt}} dB_t. \quad (64)$$

The continuous version of DDPM

Recall that

$$x_t = \sqrt{1 - \beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon. \quad (65)$$

Similarly

$$x_t \approx (1 - \frac{1}{2}\beta_t)x_{t-1} + \sqrt{\beta_t}\epsilon, \quad (66)$$

hence, we have

$$\begin{aligned} \Delta x_t &= -\frac{1}{2}\tilde{\beta}_t x_{t-1} \Delta t + \sqrt{\tilde{\beta}_t} \sqrt{\Delta t} \epsilon, \quad \text{where } \tilde{\beta}_t = \frac{\beta_t}{\Delta t} \\ &= -\frac{1}{2}\tilde{\beta}_t x_{t-1} \Delta t + \sqrt{\tilde{\beta}_t} \Delta B_t. \end{aligned} \quad (67)$$

Further, let $\Delta t \rightarrow 0$, we finally obtain

$$dx = -\frac{1}{2}\tilde{\beta}_t x dt + \sqrt{\tilde{\beta}_t} dB_t. \quad (68)$$

Summarizing the form of continuous DDPM and SMLD, we summarize the adding noise process as

$$dx = f(x, t)dt + g(t)dB_t, \quad (69)$$

where the corresponding distribution $p(x, t)$ satisfying the following Fokker-Planck equation

$$\frac{\partial p(x, t)}{\partial t} + \nabla \cdot (f(x, t)p(x, t)) - \frac{1}{2}g^2(t)\Delta p(x, t) = 0. \quad (70)$$

Training DSM object

$$J_{DSM} = \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{p(x_0)} \mathbb{E}_{p(x_t|x_0)} [\|s_\theta(x_t, t) - \nabla_{x_t} \log p(x_t|x_0)\|^2]. \quad (71)$$

VE & VP adding noise

To training the Denoising Score Match loss J_{DSM} above, the key is to add noise by the transition kernel

$$p(\tilde{x}|x) \equiv p(x(t)|x(0)) = N(x(t); \tilde{\alpha}_t x(0), \tilde{\sigma}_t^2), \quad (72)$$

from which we can obtain the noised data

$$x(t) = \tilde{\alpha}_t x(0) + \tilde{\sigma}_t \epsilon, \quad \epsilon \sim N(0, 1). \quad (73)$$

The question becomes: Given VE & VP

$$dx = \sqrt{\frac{d[\sigma^2]}{dt}} dB_t, \quad dx = -\frac{1}{2} \tilde{\beta}_t x dt + \sqrt{\tilde{\beta}_t} dB_t. \quad (74)$$

how to derive $p(x(t)|x(0))$, i.e., $p(\tilde{x}|x)$ for adding noise?

VE & VP Transition kernel

The summarizing form of VE & VP is

$$dx = h(t)xdt + g(t)dB_t, \quad (75)$$

where $h(t) = 0$ for VE and $h(t) = -\frac{1}{2}\tilde{\beta}(t)$ for VP.

Lemma

Let $\mu(t) = \mathbb{E}[x(t)]$ and $\Sigma(t) = \mathbb{E}[(x - \mu(t))(x - \mu(t))^T]$, we have the following formula

$$\frac{d\mu(t)}{dt} = h(t)\mu(t), \quad \frac{d\Sigma(t)}{dt} = 2h(t)\Sigma(t) + g^2(t)I. \quad (76)$$

The proof of the Lemma can be found in a stochastic differential equation textbook such as Applied Stochastic Differential Equations, Särkkä and Solin, 2019.

VE & VP Transition kernel

Lemma

For VE, the transition kernel is given by the following formula

$$P(x(t)|x(0)) = N(x(t); x(0), \sigma^2(t) - \sigma^2(0)). \quad (77)$$

For VE,

$$\tilde{\alpha}_t \equiv 1, \quad \tilde{\sigma}_t^2 \equiv \sigma^2(t) - \sigma^2(0) \approx \sigma^2(t). \quad (78)$$

Lemma

For VP, the transition kernel is given the following formula

$$p(x(t)|x(0)) = N\left(x(t); x(0)e^{-\frac{1}{2} \int_0^t \tilde{\beta}(s) ds}, 1 - e^{-\int_0^t \tilde{\beta}(s) ds}\right). \quad (79)$$

For VP,

$$\tilde{\alpha}_t \equiv e^{-\frac{1}{2} \int_0^t \tilde{\beta}(s) ds}, \quad \tilde{\sigma}_t^2 \equiv 1 - e^{-\int_0^t \tilde{\beta}(s) ds}. \quad (80)$$

Proof of VE transition kernel

Proof.

For VE, by a simple calculation we have

$$\frac{d\mu(t)}{dt} = 0, \quad \frac{d\Sigma(t)}{dt} = \frac{d[\sigma^2(t)]}{dt}, \quad (81)$$

Hence, it is easy to obtain the following form

$$\mu(t) = \mu(0) = x(0), \quad \Sigma(t) = \sigma^2(t) - \sigma^2(0). \quad (82)$$

Here we have used the fact

$$\mu(0) = x(0), \quad \Sigma(0) = 0 \quad (83)$$

as $x(0)$ is taken as a constant, not a random variable. And we obtain

$$P(x(t)|x(0)) = N(x(t); x(0), \sigma^2(t) - \sigma^2(0)). \quad (84)$$

VP Proof: Differential Equations

Proof.

For VP, we have the differential equations:

$$\frac{d\mu(t)}{dt} = -\frac{1}{2}\beta(t)\mu(t) \quad (85)$$

$$\frac{d\Sigma(t)}{dt} = -\beta(t)\Sigma(t) + \beta(t) \quad (86)$$

We solve these using integrating factors. □

VP Proof: Integrating Factors

Proof (Cont).

For the mean equation, multiply by $e^{\int_0^t \frac{1}{2}\beta(s)ds}$:

$$\frac{d\mu(t)}{dt} e^{\int_0^t \frac{1}{2}\beta(s)ds} + \frac{1}{2}\beta(t)\mu(t)e^{\int_0^t \frac{1}{2}\beta(s)ds} = 0 \quad (87)$$

For the variance equation, multiply by $e^{\int_0^t \beta(s)ds}$:

$$\frac{d\Sigma(t)}{dt} e^{\int_0^t \beta(s)ds} + \beta(t)\Sigma(t)e^{\int_0^t \beta(s)ds} = \beta(t)e^{\int_0^t \beta(s)ds} \quad (88)$$

Proof (Cont).

This gives us:

$$\frac{d}{dt}(\mu(t)e^{\int_0^t \frac{1}{2}\beta(s)ds}) = 0 \quad (89)$$

$$\frac{d}{dt}(\Sigma(t)e^{\int_0^t \beta(s)ds}) = \frac{d}{dt}e^{\int_0^t \beta(s)ds} \quad (90)$$

Solving with initial conditions $\mu(0) = x(0)$, $\Sigma(0) = 0$:

$$p(x(t)|x(0)) = N\left(x(t); x(0)e^{-\frac{1}{2}\int_0^t \beta(s)ds}, 1 - e^{-\int_0^t \beta(s)ds}\right) \quad (91)$$

Revisit adding noise

The **equivalence** of the adding noise

- ① From the stochastic process, denote the variable x_t , we have

$$dx_t = f(x, t)dt + g(t)dB_t, \quad (92)$$

with corresponding discrete form (Euler-Maruyama scheme)

$$x_{t+\Delta t} = x_t + f(x_t, t)\Delta t + g(t)\sqrt{\Delta t}\epsilon, \quad \epsilon \sim N(0, 1). \quad (93)$$

- ② From transition kernel perspective,

$$p(x_t) = \int_{x_0} p(x_0)p(x_t|x_0)dx_0, \quad (94)$$

where

$$p(x_t|x_0) = N(x_t; \mu_t, \Sigma_t) = N(x_t; \tilde{\alpha}_t x_0, \tilde{\sigma}_t^2) = \frac{1}{C} e^{-\frac{\|x_t - \mu_t\|^2}{2\Sigma_t}}.$$

Training recipe

The training for VE follows the following process (VP is in a similar way)

- 1 Random choose one data $x \sim p_{data}$;
- 2 Random sample $t \sim U[0, 1]$;
- 3 Random sample a white noise $\epsilon \sim N(0, 1)$;
- 4 Adding noise

$$x_t = \tilde{\alpha}_t x + \tilde{\sigma}_t \epsilon, \quad (96)$$

where the mean value $\tilde{\alpha}_t x = x(0)$ and standard deviation $\tilde{\sigma}_t = \sqrt{\sigma^2(t) - \sigma^2(0)} \approx \sigma(t)$ in VE referring to (77). VP is similar with the mean and standard deviation from its transition kernel (79);

- 5 Compute the Loss by summation the following norm over x , t and ϵ

$$\|s_\theta(x_t, t) - \epsilon / \tilde{\sigma}_t\|.$$

Inference: Reverse denoising process

The reverse denoising process is given by

$$dx = (f(x, t) - g^2 \nabla_x \log p(x, t))dt + g(t)dB_t, \quad (98)$$

or

$$dx = (f(x, t) - \frac{1}{2}g^2 \nabla_x \log p(x, t))dt, \quad (99)$$

or

$$dx = (f(x, t) - \frac{3}{2}g^2 \nabla_x \log p(x, t))dt + \sqrt{2}g(t)dB_t, \quad (100)$$

$$\dots \quad (101)$$

Due to obeying the same Fokker-Planck equation for $p(x, t)$

$$\frac{\partial p(x, t)}{\partial t} + \nabla \cdot (f(x, t)p(x, t)) - \frac{1}{2}g^2(t)\Delta p(x, t) = 0.$$

Reverse Process Proof: Setup

We prove the first reverse sampling form (98). The Fokker-Planck equation of the forward process (69) is:

$$\frac{\partial p}{\partial t} + \nabla \cdot (f(x, t)p) - \frac{1}{2}g^2 \Delta p = 0. \quad (103)$$

Let $t = T - \tau$, we have:

$$\frac{\partial p}{\partial \tau} - \nabla \cdot (f(x, T - \tau)p) + g^2 \Delta p - \frac{1}{2}g^2 \Delta p = 0. \quad (104)$$

Reverse Process Proof: Derivation

Proof (Cont).

By $\nabla \cdot \nabla = \Delta$ and $\nabla \log p = \nabla p / p$:

$$\frac{\partial p}{\partial \tau} - \nabla \cdot \left((f(x, T - \tau) - g^2 \nabla \log p) p \right) - \frac{1}{2} g^2 \Delta p = 0. \quad (105)$$

Hence, we obtain the corresponding SDE form:

$$dx = -(f(x, T - \tau) - g^2 \nabla \log p) d\tau + g dB_\tau \quad (106)$$

By substitution $\tau = T - t$:

$$dx = (f(x, t) - g^2 \nabla \log p) dt + g dB_t \quad (107)$$

Inference recipe

The inference for VE follows the following process (VP is in a similar way)

- 1 Random generate a noise at time $t = 1$ as $x(1) \sim N(0, \sigma_{max}^2)$;
- 2 Integrate the reverse sampling equation

$$dx = (f(x, t) - g^2(t)s_\theta(x, t)) dt + g(t)dB_t \quad (108)$$

or

$$dx = (f - \frac{1}{2}g^2s_\theta)dt \quad (109)$$

over the time span $[0, 1]$ to get $x(0)$.

- 3 Corrector process: at each internal value $x(t)$, we can relax the value $x(t)$ by a corrector, which takes the Langevin MCMC process.

The Corrector: revisit Langevin MCMC

The Langevin diffusion process is given by

$$x_i = x_{i-1} + \frac{\Delta t}{2} \nabla_x \log p(x_{i-1}) + \sqrt{\Delta t} \epsilon, \quad \epsilon \sim N(0, 1). \quad (110)$$

We can prove $\pi(x)$ of the obtained data points $\{x_i\}_{i=1}^N$ converges to $p(x)$.

Proof.

The continuous form of the scheme is:

$$dx = \frac{1}{2} \nabla \log p(x) dt + dB_t. \quad (111)$$

The corresponding Fokker-Planck equation is:

$$\frac{\partial \pi(x, t)}{\partial t} + \nabla \cdot \left(\frac{1}{2} \nabla \log p(x) \pi(x, t) \right) - \frac{1}{2} \Delta \pi(x, t) = 0. \quad (112)$$

With sufficient time evolution, the distribution converges to a stable distribution where:

$$\frac{\partial \pi(x, t)}{\partial t}$$

The Corrector: revisit Langevin MCMC

Proof (Cont).

Hence we have:

$$\nabla \cdot (\pi \nabla \log p - \nabla \pi) = 0, \quad \forall x. \quad (114)$$

Therefore:

$$\nabla \log p = \nabla \log \pi \quad (115)$$

given that $\pi > 0$ almost everywhere. Hence:

$$\pi^*(x) = p(x) \quad (116)$$

where π^* is the stable distribution of $\pi(x, t)$.

Outline

- 1 Preliminary: Generating Samples from Probability Distributions
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- 5 Diffusion model: from theories to implementation
- 6 Continuous version of Diffusion Model
- 7 Mathematical motivation for Flow Matching**
- 8 Connection with Diffusion Models
- 9 Conditional Paths and Velocity Targets
- 10 Flow Matching Training and Inference

Probability Path and Coupling

- Let $\{p_t\}_{t \in [0,1]}$ be a prescribed path of marginals on $\mathbb{R}^d$ with p_0 simple and p_1 the data.
- A *coupling* induces random trajectories $t \mapsto x_t$ with instantaneous velocity $\dot{x}_t = \frac{d}{dt}x_t$ (may be stochastic given x_t).
- Goal: learn a *deterministic* vector field $v : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ such that the ODE

$$\frac{d}{dt}x_t = v(x_t, t), \quad x_0 \sim p_0,$$

yields marginals $x_t \sim p_t$ for all t .

Remark

We can depict the flow matching and diffusion through the probability transition v.s. the trajectories. And also, we can connect the Neural ODE with the Flow matching.

Strong and Weak Continuity

From the Fokker–Planck equation, for $\frac{dx}{dt} = v(x, t)$, we have

$$\partial_t p_t(x) + \nabla \cdot (p_t(x) v(x, t)) = 0.$$

The corresponding weak form of above strong form is

Weak form (for any test functions $\varphi \in C_c^\infty(\mathbb{R}^d)$)

$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x) p_t(x) dx = \int_{\mathbb{R}^d} \nabla \varphi(x) \cdot v(x, t) p_t(x) dx,$$

i.e.,

$$\frac{d}{dt} \mathbb{E}[\varphi(x_t)] = \mathbb{E}[\nabla \varphi(x_t) \cdot v(x, t)].$$

Weak Form from Random Trajectories

Let x_t be the coupled trajectory (not necessarily deterministic). For any test function $\phi \in C_c^\infty(\mathbb{R}^d)$, we have

$$\frac{d}{dt} \mathbb{E}[\varphi(x_t)] = \mathbb{E}[\nabla \varphi(x_t) \cdot \dot{x}_t],$$

by the expectation of the derivative of the test function.

Expanding the expectation into integrals at a fixed t :

$$\begin{aligned} \mathbb{E}[\nabla \varphi(x_t) \cdot \dot{x}_t] &= \iint_{\mathbb{R}^d \times \mathbb{R}^d} \nabla \varphi(x) \cdot \dot{x} \mu_t(dx, d\dot{x}) \\ &= \int_{\mathbb{R}^d} \nabla \varphi(x) \cdot \left(\int \dot{x} p(\dot{x} | x, t) d\dot{x} \right) p_t(x) dx \\ &= \int_{\mathbb{R}^d} \nabla \varphi(x) \cdot v^*(x, t) p_t(x) dx = \mathbb{E}[\nabla \varphi(x_t) \cdot v^*(x, t)]. \end{aligned}$$

where $v^*(x, t) \equiv \mathbb{E}[\dot{x}_t | x_t = x, t] = \int_{\mathbb{R}^d} \dot{x} p(\dot{x} | x, t) d\dot{x}$. Thus the random path induces the weak identity with $v = v^*$.

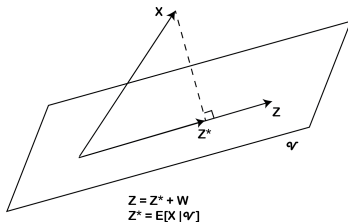
Theoretical Support: Conditional Expectation

Theorem

Let X be an integrable random variable. Then for each σ -algebra $\mathcal{V}$ and $Z \in \mathcal{V}$, $Z^* = \mathbb{E}(X|\mathcal{V})$ solves the least square problem

$$\operatorname{argmin}_{Z \in \mathcal{V}} \|Z - X\|,$$

where $\|Z\| = \left(\int Z^2 dP\right)^{\frac{1}{2}}$.



Hence we have $L = \mathbb{E}\|v_\theta(x_t, t) - \dot{x}_t\|^2$.

Practical Training Recipe (Per-Iteration)

- 1 Sample $t \sim w(t)$ (typically uniform on $[0, 1]$).
- 2 Sample $(x_t, \dot{x}_t)$ from the chosen path coupling.
- 3 Regress $\dot{x}_t$ on (x_t, t) via the MSE:

$$\mathcal{L} = \|v_\theta(x_t, t) - \dot{x}_t\|^2.$$

- 4 Update θ by SGD; no density estimation, no reverse simulation is needed during training.

Remark

Different path/coupling choices change v^ and the irreducible variance $\mathbb{E}\|\dot{x}_t - v^*\|^2$, affecting sample efficiency and optimization, but the optimal target remains the conditional mean.*

Key Takeaways

- The correct notion of “matching the path” is the **weak continuity** identity for all test functions.
- The only deterministic field depending on (x, t) that preserves this identity is the **conditional mean velocity**

$$v^*(x, t) = \mathbb{E}[\dot{x}_t \mid x_t = x, t] .$$

- The **flow-matching MSE** is precisely the L^2 projection selecting v^* :

$$v_\theta \xrightarrow[\text{minimize}]{\mathbb{E} \|v_\theta(x_t, t) - \dot{x}_t\|^2} v^* .$$

Outline

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- 3 Relations of different models: x_0 , epsilon and score model
- 4 Potential Score Matching loss
- 5 Diffusion model: from theories to implementation
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- 8 Connection with Diffusion Models**
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Theorem (Fokker–Planck / Forward Kolmogorov)

Let $(X_t)_{t \geq 0} \in \mathbb{R}^N$ solve the Itô SDE

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t,$$

where $\mu : \mathbb{R}^N \times [0, \infty) \rightarrow \mathbb{R}^N$ is the drift, $\sigma : \mathbb{R}^N \times [0, \infty) \rightarrow \mathbb{R}^{N \times M}$ the diffusion, and W_t an M -dimensional standard Wiener process. Define the diffusion tensor

$$D(x, t) := \frac{1}{2} \sigma(x, t) \sigma(x, t)^\top \in \mathbb{R}^{N \times N}.$$

Then the (smooth) density $p(x, t)$ satisfies the Fokker–Planck equation

$$\partial_t p(x, t) = - \sum_{i=1}^N \frac{\partial}{\partial x_i} [\mu_i(x, t) p(x, t)] + \sum_{i,j=1}^N \frac{\partial^2}{\partial x_i \partial x_j} [D_{ij}(x, t) p(x, t)]$$

Isotropic diffusion: $\sigma(x, t) = \sigma(t) \mathbf{I}_N$

When $\sigma(x, t) = \sigma(t) \mathbf{I}_N$, the diffusion tensor becomes

$$D(x, t) = \frac{1}{2} \sigma(t)^2 \mathbf{I}_N, \quad D_{ij}(x, t) = \frac{1}{2} \sigma(t)^2 \delta_{ij}.$$

Hence the Fokker–Planck equation simplifies to the drift–diffusion form

$$\partial_t p(x, t) = -\nabla \cdot (\mu(x, t) p(x, t)) + \frac{\sigma(t)^2}{2} \Delta p(x, t)$$

where $\nabla \cdot (\mu p) = \sum_{i=1}^N \partial_{x_i} (\mu_i p)$ and $\Delta = \sum_{i=1}^N \partial_{x_i}^2$ is the Laplacian.

Mathematical connection to diffusion models

Theoretical link: For a forward SDE $d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) dB_t$, the probability flow ODE is:

$$\frac{d}{dt} \mathbf{x} = f(\mathbf{x}, t) - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p(\mathbf{x}, t). \quad (117)$$

Key mathematical insight: When Flow Matching uses Gaussian paths that match the SDE's marginal distributions p_t , the learned velocity field $\mathbf{v}_\theta$ approximates the probability flow drift:

$$\mathbf{v}_\theta(\mathbf{x}, t) \approx f(\mathbf{x}, t) - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p(\mathbf{x}, t) \quad (118)$$

Crucial advantage: Flow Matching achieves this approximation *without* ever computing the score $\nabla_{\mathbf{x}} \log p(\mathbf{x}, t)$ during training!

Practical implications: This connection allows Flow Matching to inherit the theoretical guarantees of diffusion models while avoiding their computational bottlenecks.

Mathematical derivation: probability flow ODE

Setup: Forward SDE $d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) dB_t$ with density $p(\mathbf{x}, t)$ solving the Fokker-Planck equation:

$$\partial_t p = -\nabla \cdot (fp) + \frac{1}{2}g^2 \Delta p. \quad (119)$$

Key insight: A deterministic ODE $\dot{\mathbf{x}} = \mathbf{v}(\mathbf{x}, t)$ transports densities via the continuity equation:

$$\partial_t p + \nabla \cdot (p\mathbf{v}) = 0. \quad (120)$$

Mathematical derivation: Equating the two equations:

$$-\nabla \cdot (p\mathbf{v}) = -\nabla \cdot (fp) + \frac{1}{2}g^2 \Delta p \quad (121)$$

$$\Rightarrow \mathbf{v} = f - \frac{1}{2}g^2 \frac{\nabla p}{p} = f - \frac{1}{2}g^2 \nabla \log p. \quad (122)$$

Result: The probability flow ODE drift is $\mathbf{v} = f - \frac{1}{2}g^2 \nabla \log p$.

Connection to Flow Matching: When Flow Matching learns $\mathbf{v}_\theta$, it approximates this probability flow drift without computing the score $\nabla \log p$.

Questions

From

- ① Score model $s_\theta(x_t) \rightarrow s_{\theta^*}(x_t) \equiv \mathbb{E}[\nabla \log p(x_t|x_0) | x_t]$
- ② X prediction model (denoising model, x_0 model) $x_\theta(x_t) \rightarrow x_{\theta^*}(x_t) \equiv \mathbb{E}[x_0 | x_t]$;
- ③ ϵ prediction model $\epsilon_\theta(x_t) \rightarrow \epsilon_{\theta^*}(x_t) \equiv \mathbb{E}[\epsilon | x_t]$.

and

$$v^*(x_t) = \mathbb{E}[\dot{x}_t | x_t]$$

, what is the relationship between optimal $v_\theta(x, t)$ and the above models? Hints:

$$v_{\theta^*}(x_t) = \mathbb{E}[\dot{x}_t | x_t] = \mathbb{E}[\dot{\alpha}(t)x_t + \dot{\sigma}(t)\epsilon | x_t]. \quad (123)$$

and

$$s(x_t) = \mathbb{E}_x[\nabla_{x_t} \log p(x_t|x_0)|x_t] = \mathbb{E}_x[-\frac{x_t - \alpha x_0}{\sigma^2} | x_t] = -\frac{x_t - \alpha \mathbb{E}[x_0|x_t]}{\sigma^2}. \quad (124)$$

Outline

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- 3 Relations of different models: x_0 , epsilon and score model
- 4 Potential Score Matching loss
- 5 Diffusion model: from theories to implementation
- 6 Continuous version of Diffusion Model
- 7 Mathematical motivation for Flow Matching
- 8 Connection with Diffusion Models
- 9 Conditional Paths and Velocity Targets**
- 10 Flow Matching Training and Inference

Gaussian conditional paths: mathematical derivation

Assumption: Gaussian conditional paths

$$q_t(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_t(\mathbf{x}_0), \sigma_t^2 \mathbf{I}), \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0) + \sigma_t \boldsymbol{\epsilon}. \quad (125)$$

Mathematical derivation: Taking the time derivative of the path:

$$\frac{d}{dt} \mathbf{x}_t = \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \dot{\sigma}_t \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} = \frac{\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_0)}{\sigma_t}. \quad (126)$$

Conditional expectation: Since $\boldsymbol{\epsilon}$ is independent of $\mathbf{x}_t$ given $\mathbf{x}_0$, we have:

$$\boxed{v = \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \frac{\dot{\sigma}_t}{\sigma_t} (\mathbf{x} - \boldsymbol{\mu}_t(\mathbf{x}_0))}. \quad (127)$$

Mathematical analysis of path choices

1. Linear bridge (optimal transport):

- $\mu_t(\mathbf{x}_0) = (1 - t) \mathbf{x}_0, \sigma_t = t \sigma_{\max}$
- **Mathematical property:** Minimizes Wasserstein-2 distance
- **Velocity:** $v = \frac{\sigma_{\max}}{t}(\mathbf{x} - \mathbf{x}_0)$
- **Advantage:** Direct connection to optimal transport theory

2. VP-like (DDPM-inspired):

- $\mu_t(\mathbf{x}_0) = \alpha(t) \mathbf{x}_0, \sigma_t = \sqrt{1 - \alpha^2(t)}$
- **Mathematical property:** Preserves energy structure
- **Velocity:** $v = \dot{\alpha}(t) \mathbf{x}_0 + \frac{\dot{\sigma}_t}{\sigma_t}(\mathbf{x} - \alpha(t) \mathbf{x}_0)$
- **Advantage:** Compatible with existing diffusion model schedules

3. VE-like (SMLD-inspired):

- $\mu_t(\mathbf{x}_0) = \mathbf{x}_0, \sigma_t = \sigma(t)$ increasing
- **Mathematical property:** Preserves data structure
- **Velocity:** $v = \frac{\dot{\sigma}(t)}{\sigma(t)}(\mathbf{x} - \mathbf{x}_0)$
- **Advantage:** Simple form, good for high-dimensional data

Key insight: All choices provide *closed-form* $\mathbf{u}_t$ via the boxed formula!

Derivation: VE/VP target forms

VE-like. $\mu_t(\mathbf{x}_0) = \mathbf{x}_0$, $\sigma_t = \sigma(t)$. Then

$$\mathbf{v} = 0 + \frac{\dot{\sigma}_t}{\sigma_t} (\mathbf{x} - \mathbf{x}_0). \quad (128)$$

VP-like. $\mu_t(\mathbf{x}_0) = \alpha(t) \mathbf{x}_0$, $\sigma_t = \sqrt{1 - \alpha^2(t)}$. Using $\mathbf{u}_t = \dot{\mu}_t + (\dot{\sigma}_t/\sigma_t)(\mathbf{x} - \mu_t)$, we get

$$\mathbf{v} = \dot{\alpha} \mathbf{x}_0 + \frac{\dot{\sigma}_t}{\sigma_t} \mathbf{x} - \frac{\dot{\sigma}_t}{\sigma_t} \alpha \mathbf{x}_0 \quad (129)$$

$$= \frac{\dot{\sigma}_t}{\sigma_t} \mathbf{x} + \left(\dot{\alpha} - \alpha \frac{\dot{\sigma}_t}{\sigma_t} \right) \mathbf{x}_0. \quad (130)$$

No explicit form for $\dot{\sigma}_t/\sigma_t$ is required to train; schedules determine it numerically.

Mathematical insight: These derivations show how different path choices lead to different velocity field structures, all computable analytically.

Outline

- 1 Preliminary: Generating Samples from Probability Distributions
- 2 Matching score for training
- 3 Relations of different models: x_0 , epsilon and score model
- 4 Potential Score Matching loss
- 5 Diffusion model: from theories to implementation
- 6 Continuous version of Diffusion Model
- 7 Mathematical motivation for Flow Matching
- 8 Connection with Diffusion Models
- 9 Conditional Paths and Velocity Targets
- 10 Flow Matching Training and Inference

Flow Matching: mathematical training objective

Theoretical foundation: Given conditional paths $q_t(\mathbf{x} \mid \mathbf{x}_0)$, the optimal velocity field satisfies:

$$\mathbf{v}^*(\mathbf{x}, t) = \mathbb{E}[\dot{\mathbf{x}}_t \mid \mathbf{x}_t = \mathbf{x}, t] \quad (131)$$

Training procedure:

- 1 Sample $t \sim \mathcal{U}[0, 1]$, $\mathbf{x}_0 \sim p_{\text{data}}$, $\mathbf{x}_t \sim q_t(\cdot \mid \mathbf{x}_0)$
- 2 Compute analytic target $\mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0)$ from chosen path
- 3 Minimize regression loss

Mathematical objective:

$$J_{\text{FM}} = \mathbb{E}[\|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{u}_t\|^2]. \quad (132)$$

Key mathematical advantage: No score estimation needed; all targets are analytic!

Computational benefits:

- Stable gradients (no exploding/vanishing)
- Parallelizable across time steps
- No special architectural requirements

Mathematical foundation of inference

Theoretical basis: The learned velocity field $\mathbf{v}_\theta(\mathbf{x}, t)$ defines a deterministic ODE:

$$\frac{d}{dt} \mathbf{x} = \mathbf{v}_\theta(\mathbf{x}, t), \quad t \in [0, 1] \quad (133)$$

Mathematical properties:

- **Existence:** Under Lipschitz conditions, solutions exist and are unique
- **Consistency:** The ODE preserves the continuity equation
- **Stability:** Deterministic integration avoids accumulation of stochastic errors

Generation process:

- 1 Initialize $\mathbf{x}(1) \sim p_1$ (e.g., $\mathcal{N}(0, \mathbf{I})$)
- 2 Integrate ODE backwards: $\frac{d}{dt} \mathbf{x} = \mathbf{v}_\theta(\mathbf{x}, t)$, $t : 1 \rightarrow 0$
- 3 Use standard ODE solvers (Euler, RK4, Dormand-Prince, etc.)

Mathematical advantages:

- Deterministic sampling (reproducible)
- Fast with large solver steps
- No stochastic correctors needed

Welcome inputs

Any comments?

Welcome your inputs!